

MSCS Final Project:

Real-Time Spike Detection Pipeline

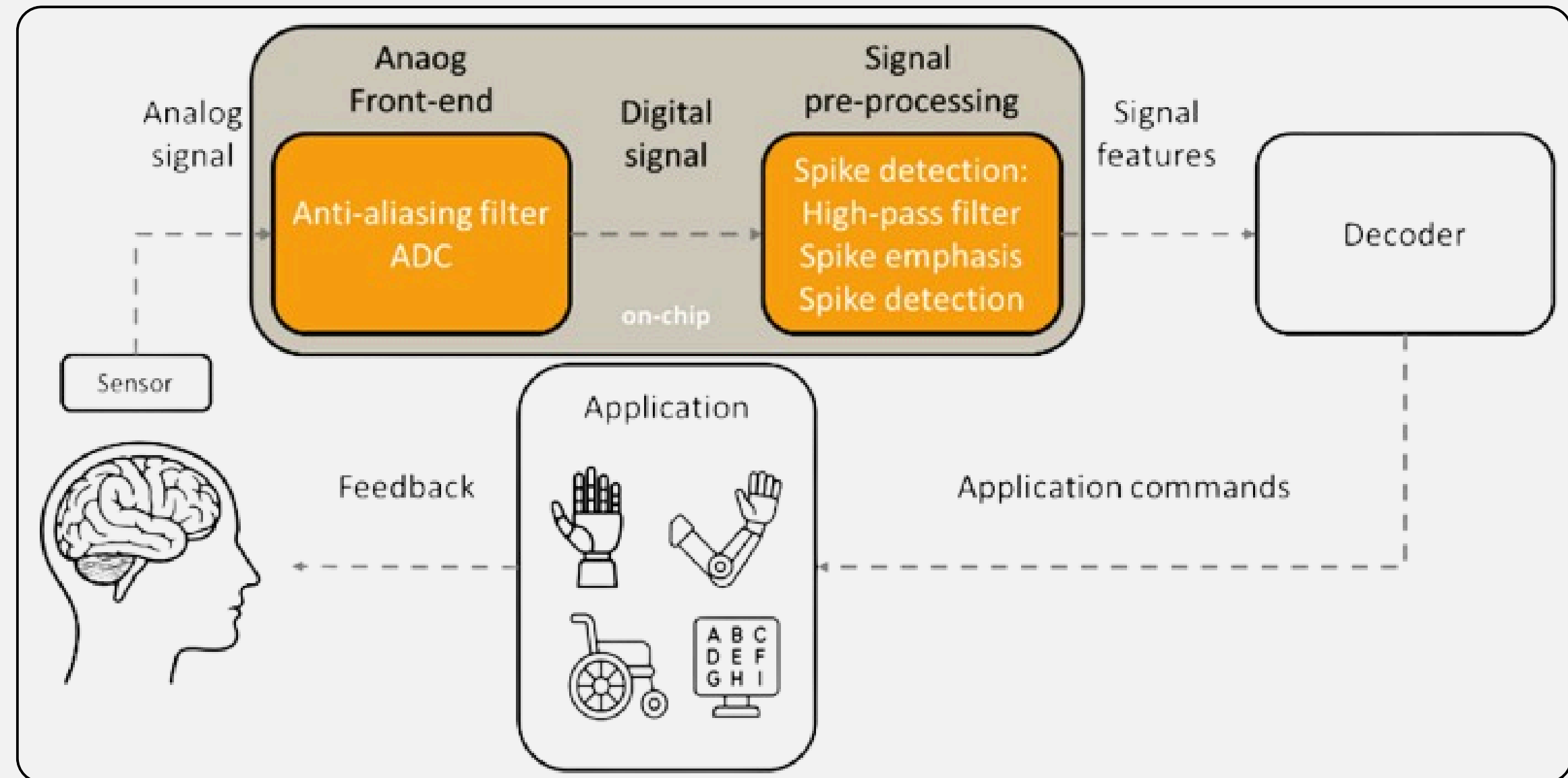
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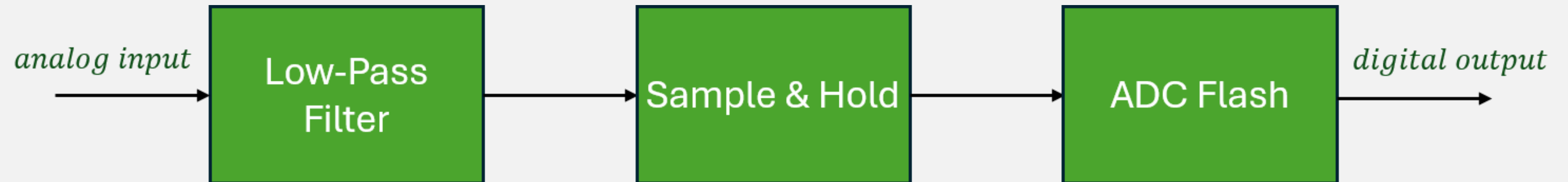


Introduction

- Objective:
Create a spike detector designing both the analog and the digital parts
- Application:
Implantable Brain - Computer Interface
- Design Focus:
High accuracy, low power consumption



Analog Front-End



1st order active
Low-Pass Filter

Sample & Hold

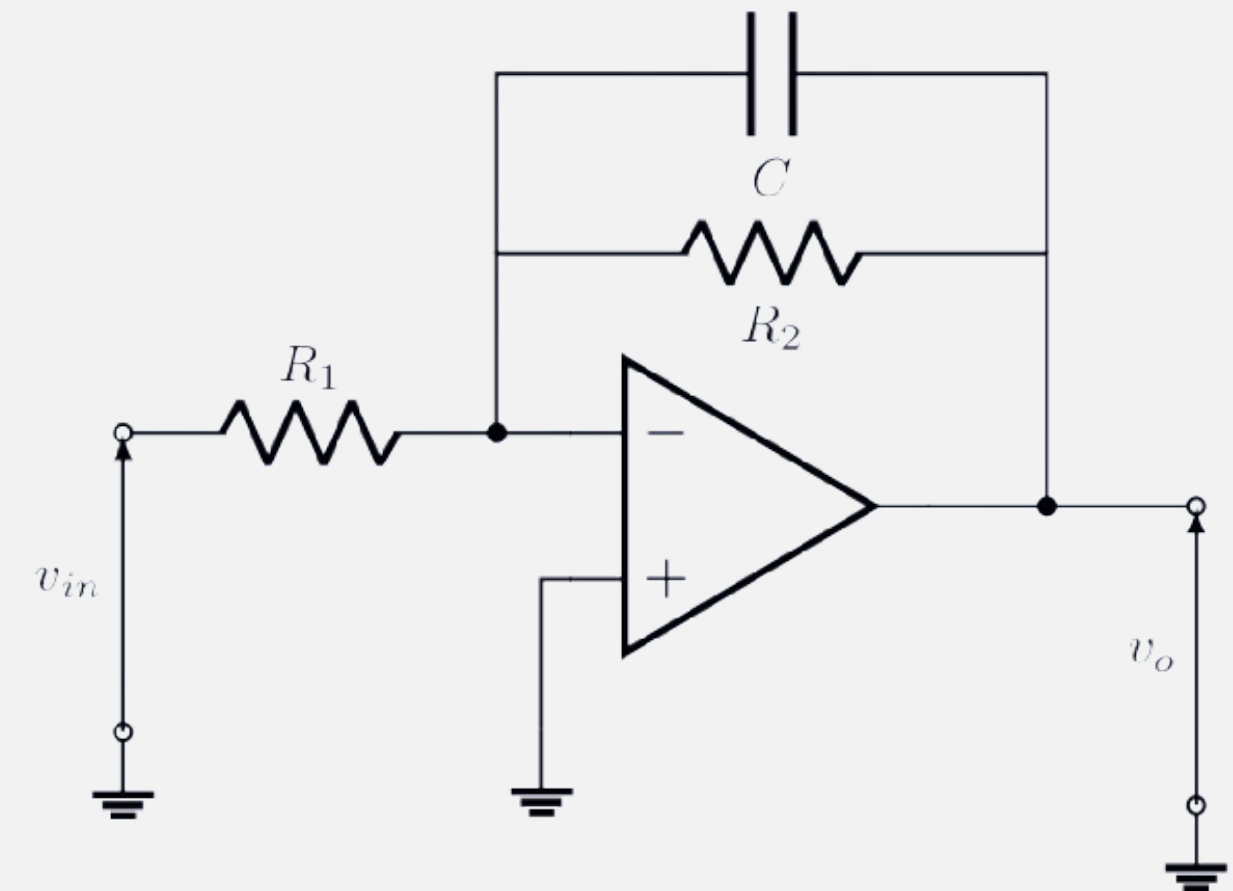
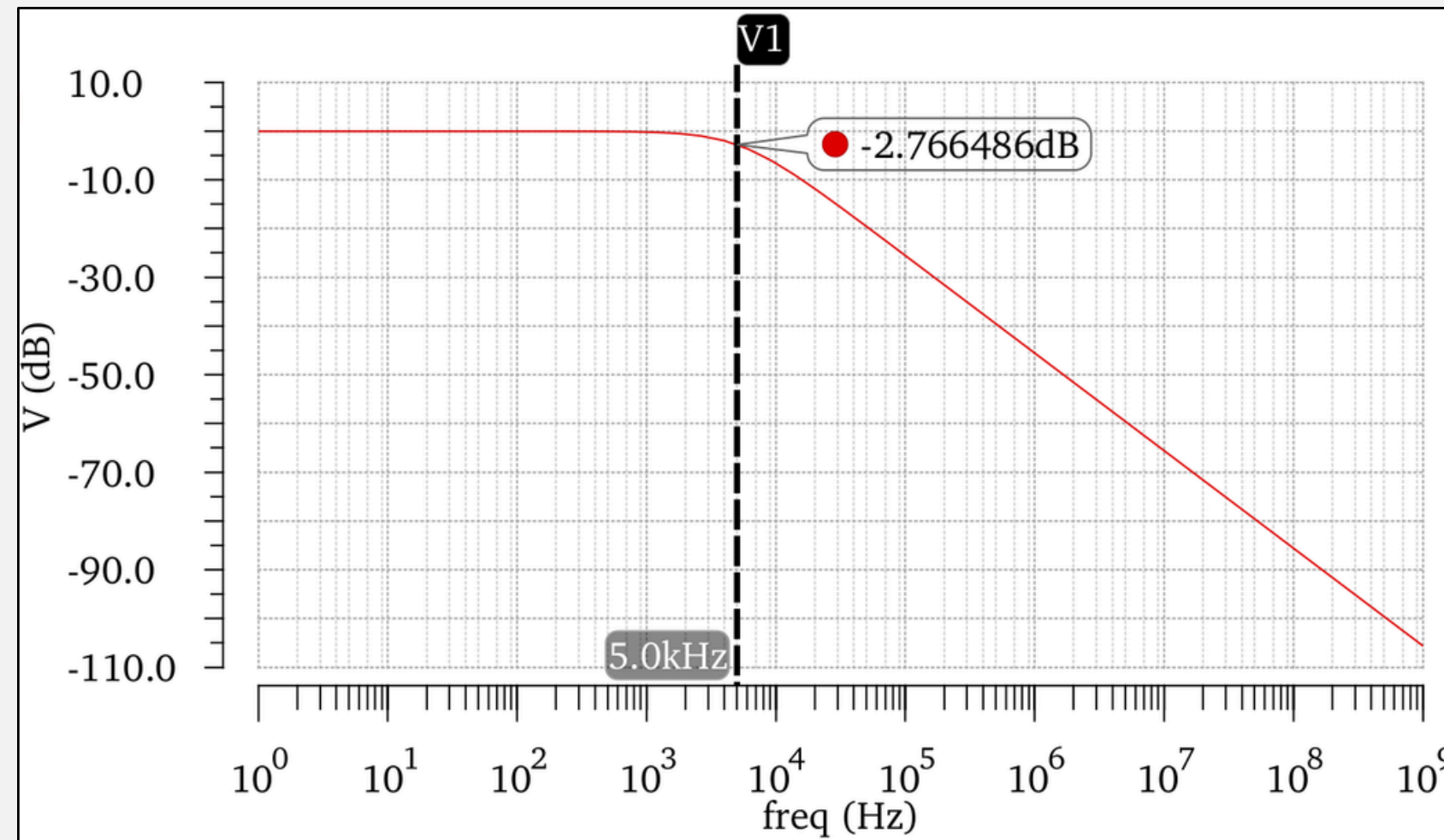
ADC flash (8 bit)

- ResNet
- CompNet
- NandNet
- Encoder

Constraints:

- Sampling frequency: 24 kHz

Anti-aliasing filter: 1st order LPF

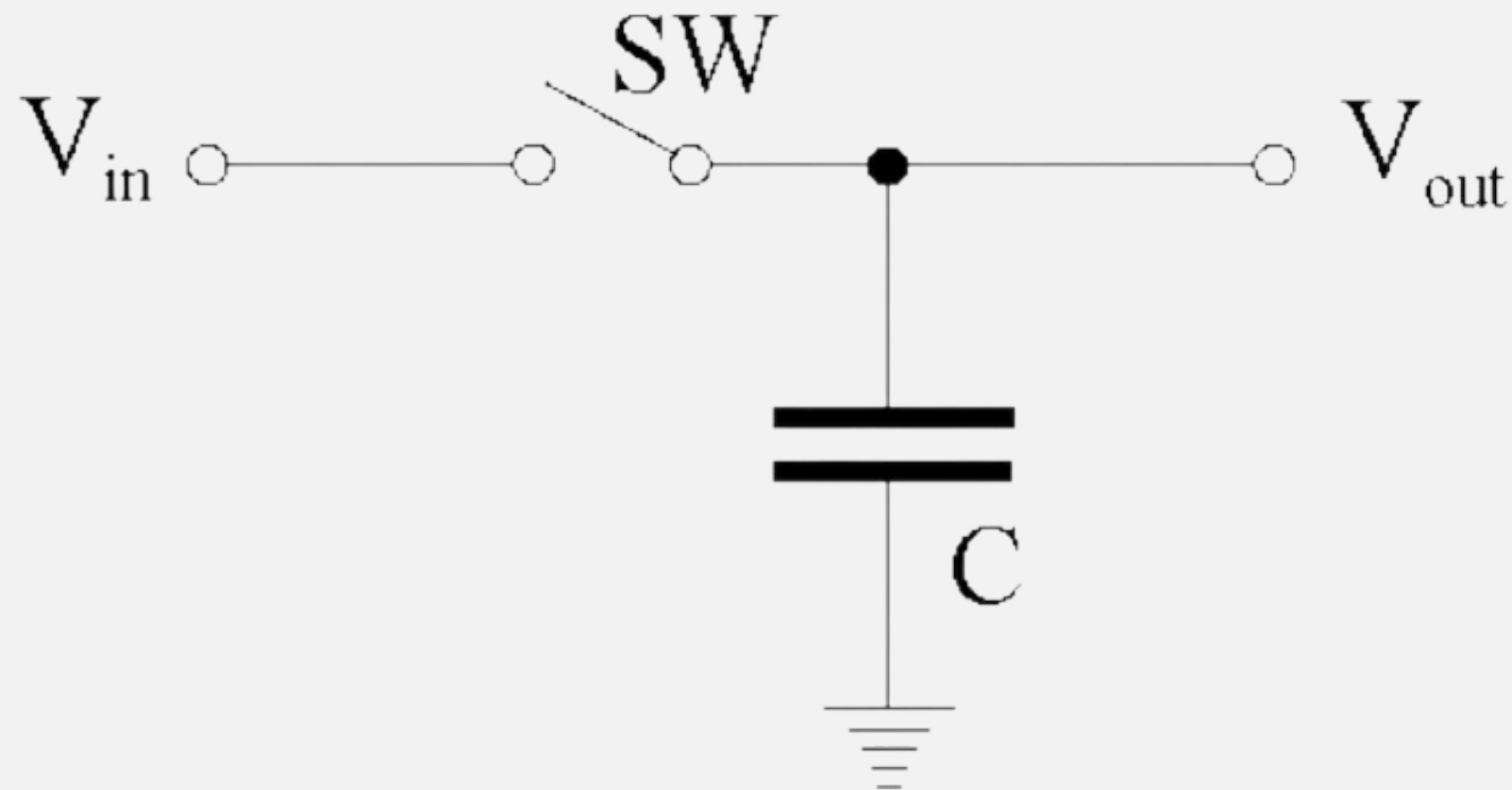


$$H(s) = -\frac{R_2}{R_1} \cdot \frac{1}{1 + s \cdot R_1 \cdot C}$$

Neural activity characteristics lies in the range of a few kHz

$$\begin{cases} f_c = \frac{1}{2\pi \cdot R \cdot C} = 5kHz \\ |A_{DC}| = \frac{R_2}{R_1} = 1 \frac{V}{V} \end{cases} \longrightarrow \begin{cases} R_1 = R_2 = 3M\Omega \\ C = 10pF \end{cases}$$

Sample & Hold Circuit



Analog constraint: $f_s = 24kHz$

The analog voltage must be stable for:

$$t = \frac{T_s}{2} = 20.83\mu s$$

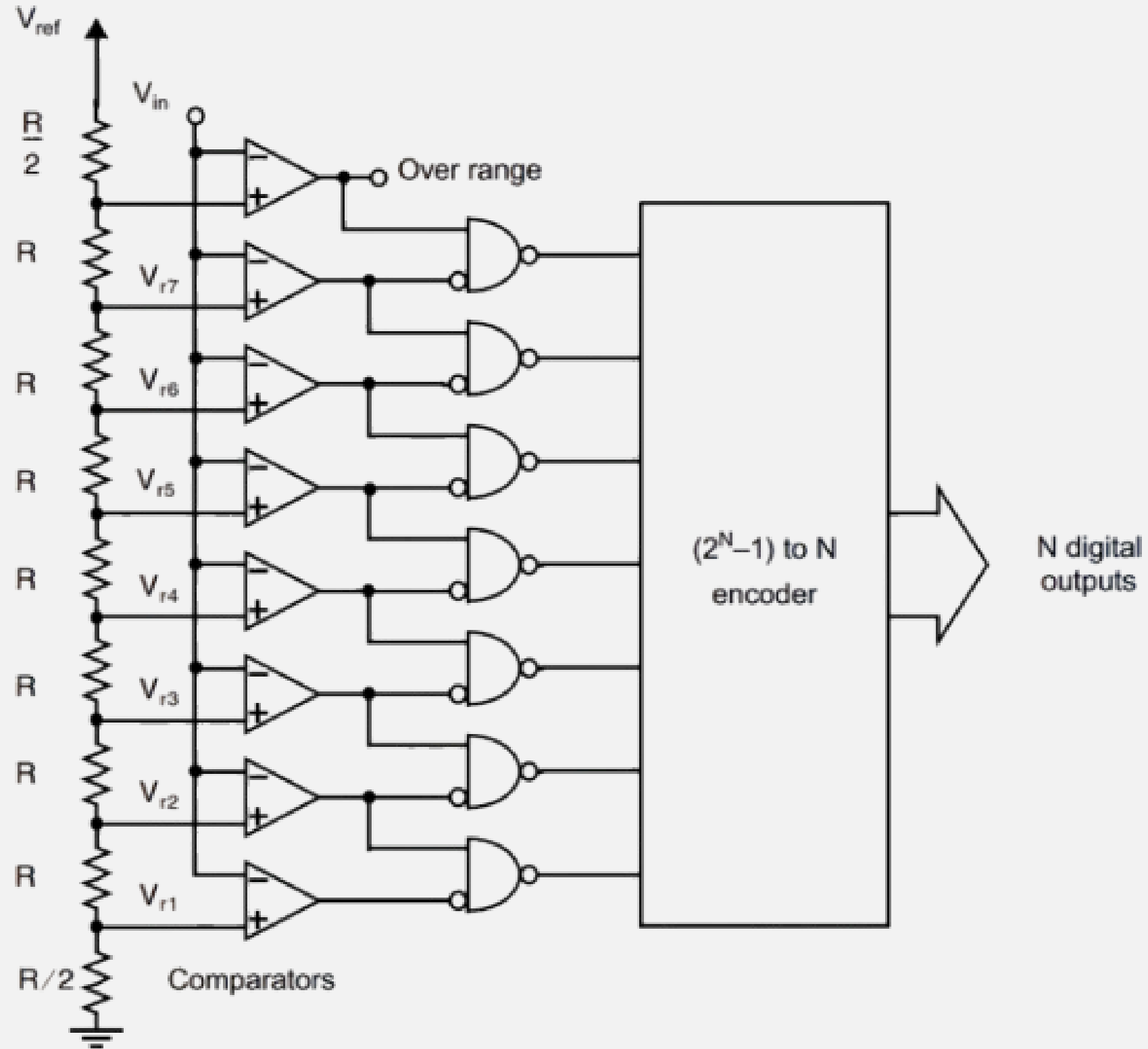
Sizing C with $R = 10\Omega$

$$\begin{cases} V_c(t) = V_f + (V_i - V_f) \cdot e^{-t/\tau} \\ V_f - V_c(t) \leq \frac{V_{LSB}}{2} \end{cases}$$

Considering: $\begin{cases} V_i = -1.5V \\ V_f = 1.5V \end{cases}$

$$\longrightarrow C_{S\&H} \leq \frac{t}{6.25 \cdot R_{S\&H}} = 339 \text{ nF} \quad \boxed{C_{S\&H} = 100 \text{ nF}}$$

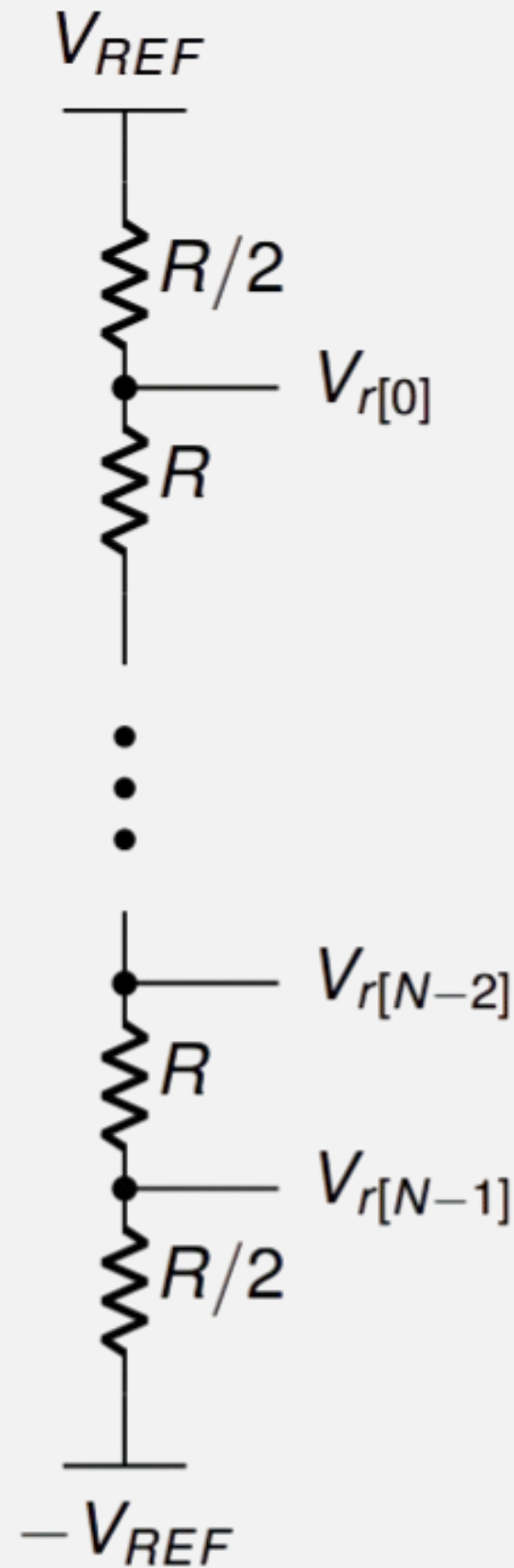
ADC Flash



Main Components and Features:

- ResNet → bipolar power supply
- CmpNet → compare input with references
- NandNet → One-Hot Encoding
- Encoder → centered output

Resistor Network



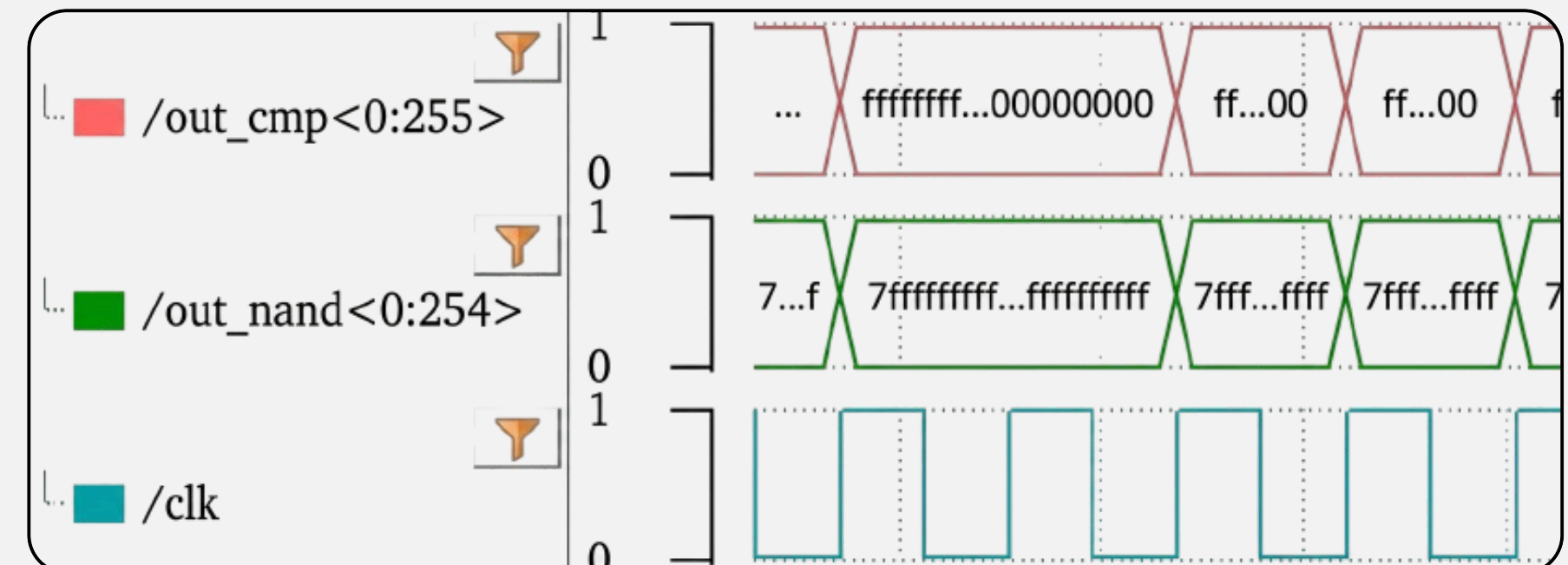
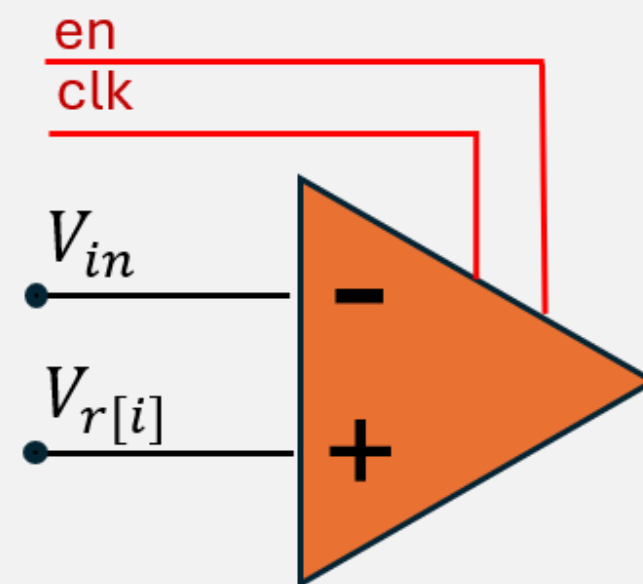
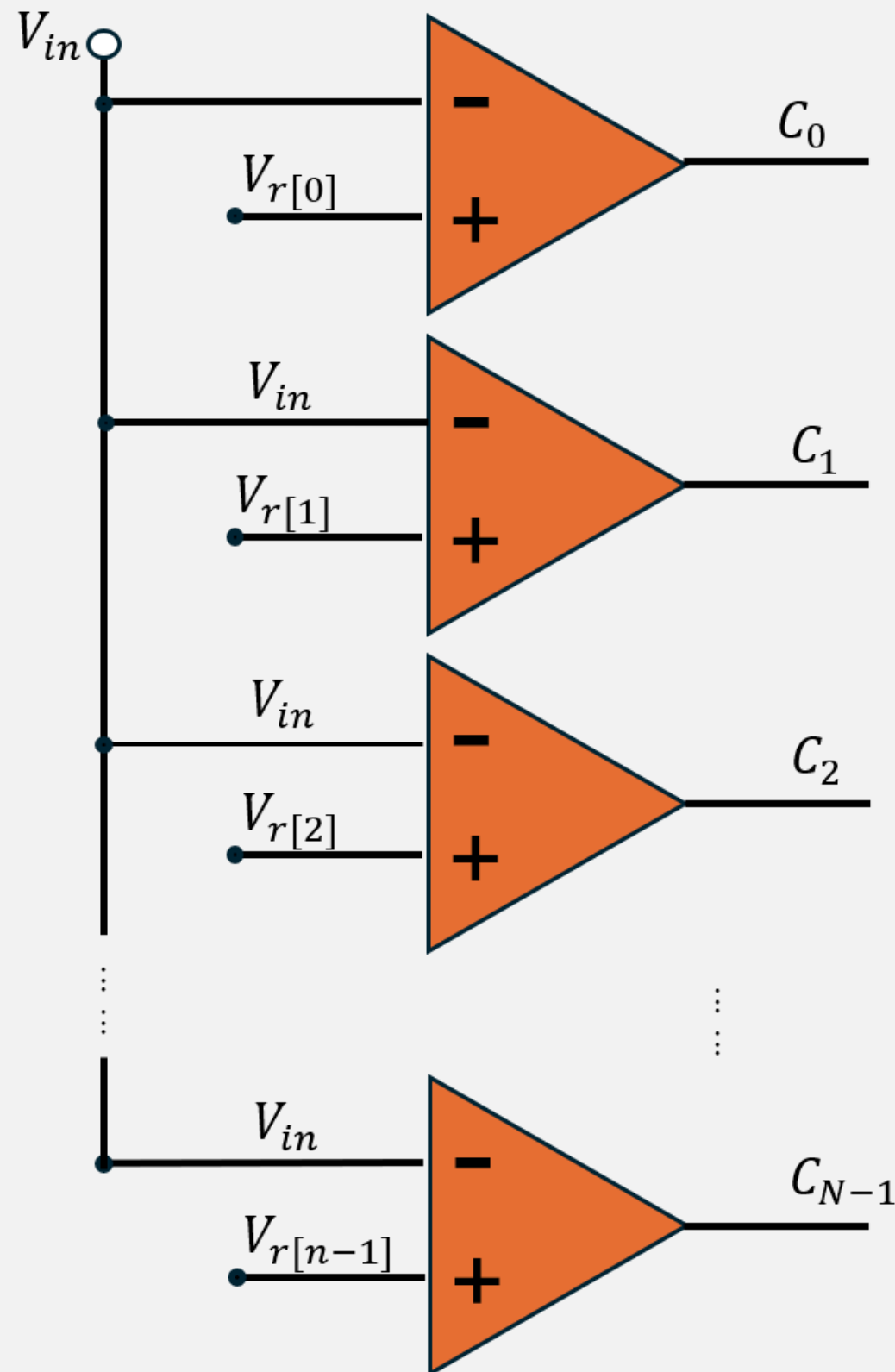
Features

- Input data is **bipolar** → need to read negative and positive values
- Maximum excursion (-1.22V , 0.88V) → $V_{REF} = 1.5V$
- $V_{LSB} = \frac{FSR}{2^8} = 11.72mV$
- First and last resistor equal to $R/2$ → **reduce quantization error**
- $R = 1\text{ k}\Omega$ → arbitrary chosen

Comparator Network

Features

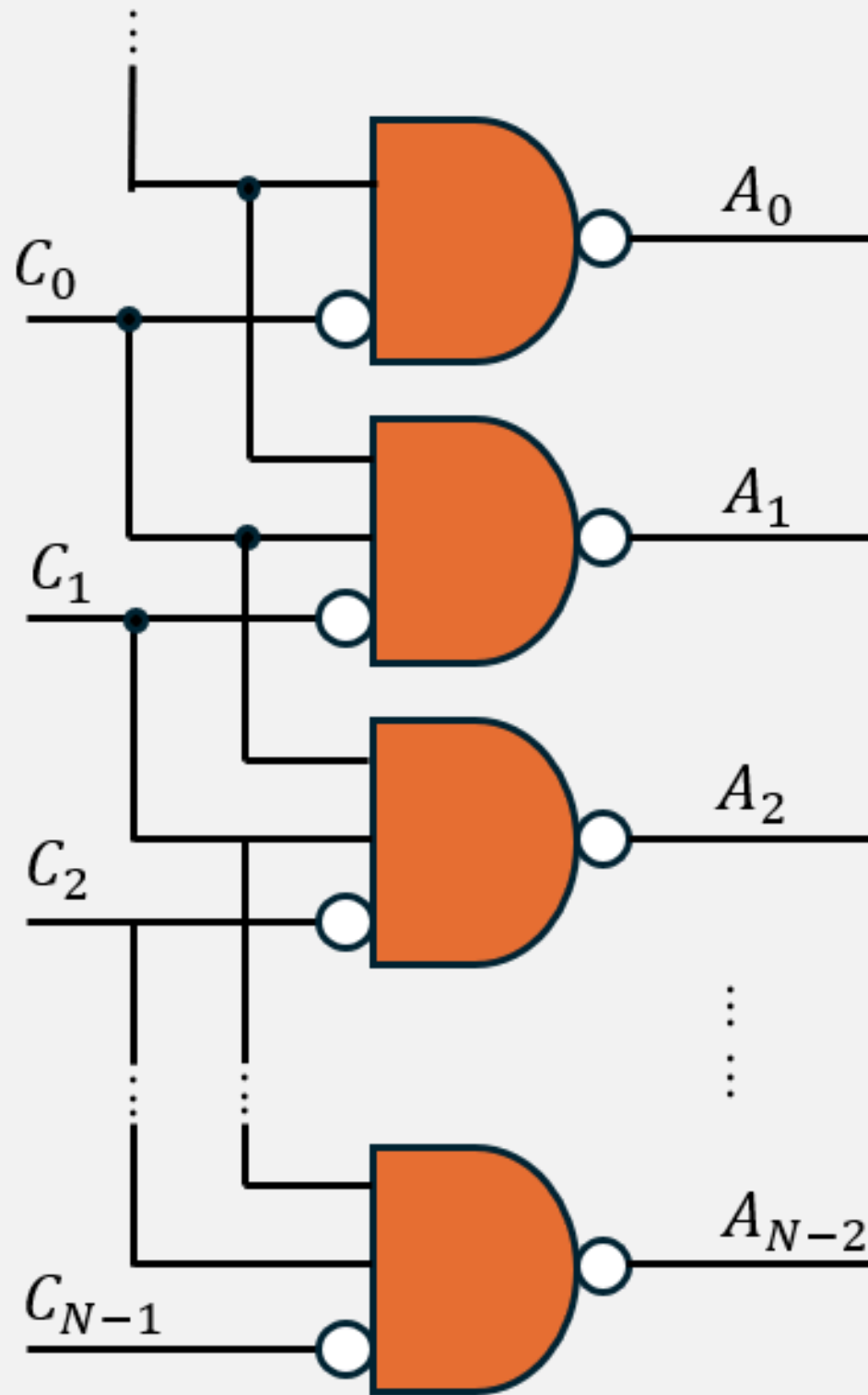
- Thermometer Encoding
- Parametric Delay $\rightarrow$ triggers bubbles
- Sequential Comparator $\rightarrow$ prevents glitches propagation



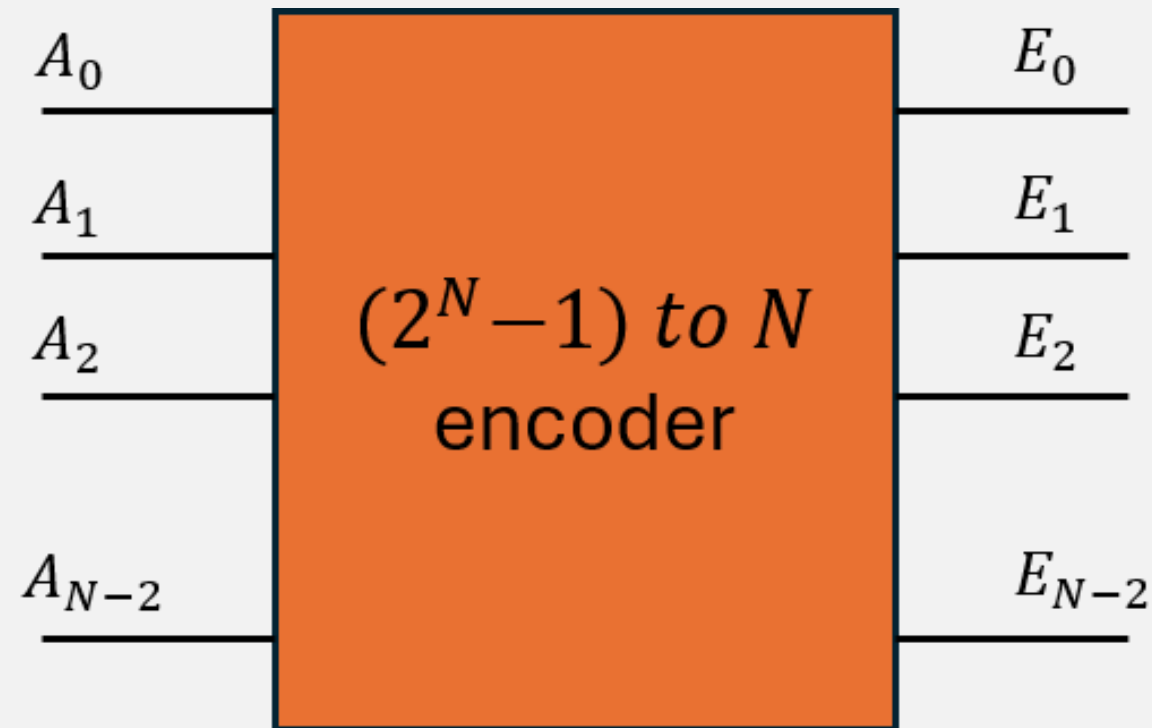
NAND Network

Features

- One Hot Encoding
- 3 input NAND → in charge to solve bubbles
- Bubble Fix → only type “1” bubbles of any order



Encoder



```
always @(*) begin
    out_enc = 0; // default value
    if(over_range == 0) out_enc = 127;
    else if(in_enc == {255{1'b1}}) out_enc = -128;
    else begin
        for (k = 0; k <= N-2; k = k + 1) begin
            if (in_enc[k] == 1'b0) begin
                out_enc = -(N-2 - k + 1 + 128);
            end
        end
    end
end
end
```

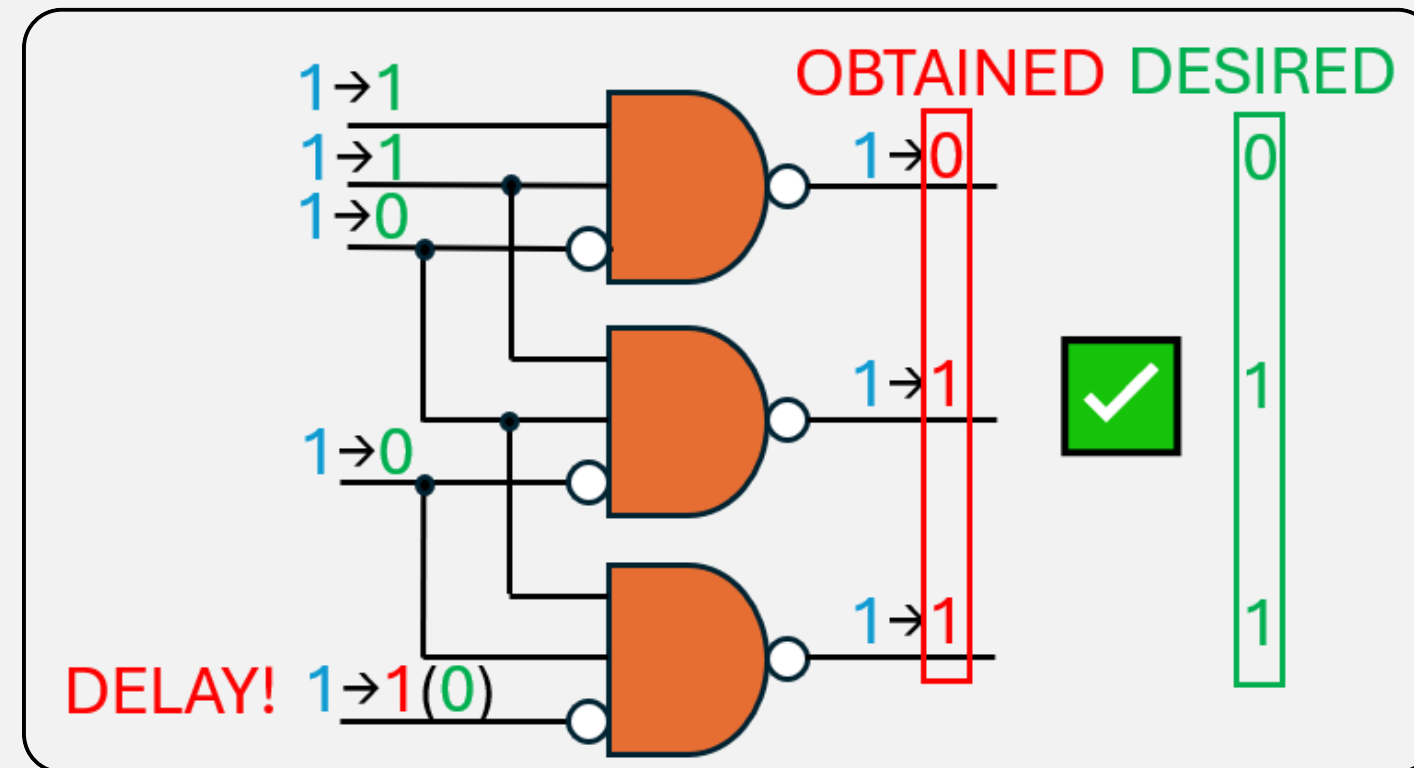
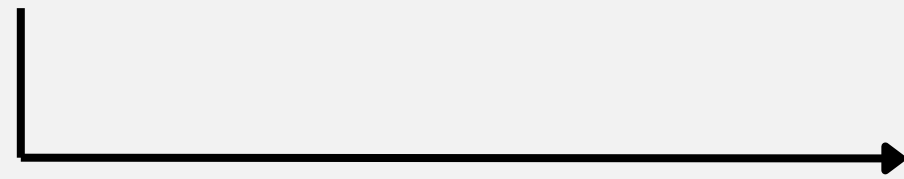
Features

- Centered output interval $\rightarrow [-128 ; 127]$
- Overvoltage correction:
 - $V_{in} < -V_{ref} \rightarrow$ saturates to -128
 - $V_{in} > V_{ref} \rightarrow$ saturates to 127
- Bubble Fix $\rightarrow$ only type “0” bubbles of order greater than 2

Bubbles

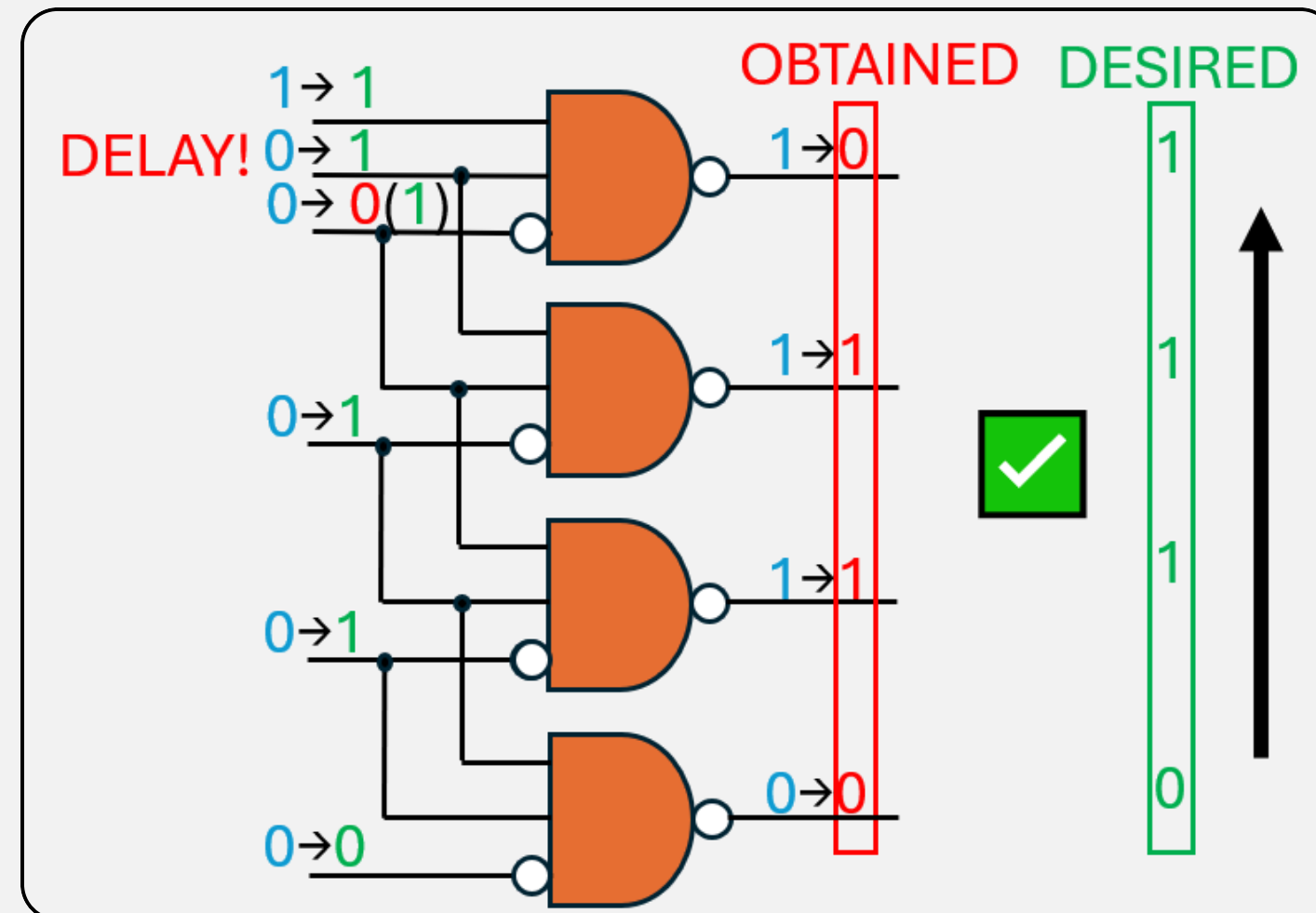
Type “1” bubbles (any order):

- Solved by the 3 inputs NAND



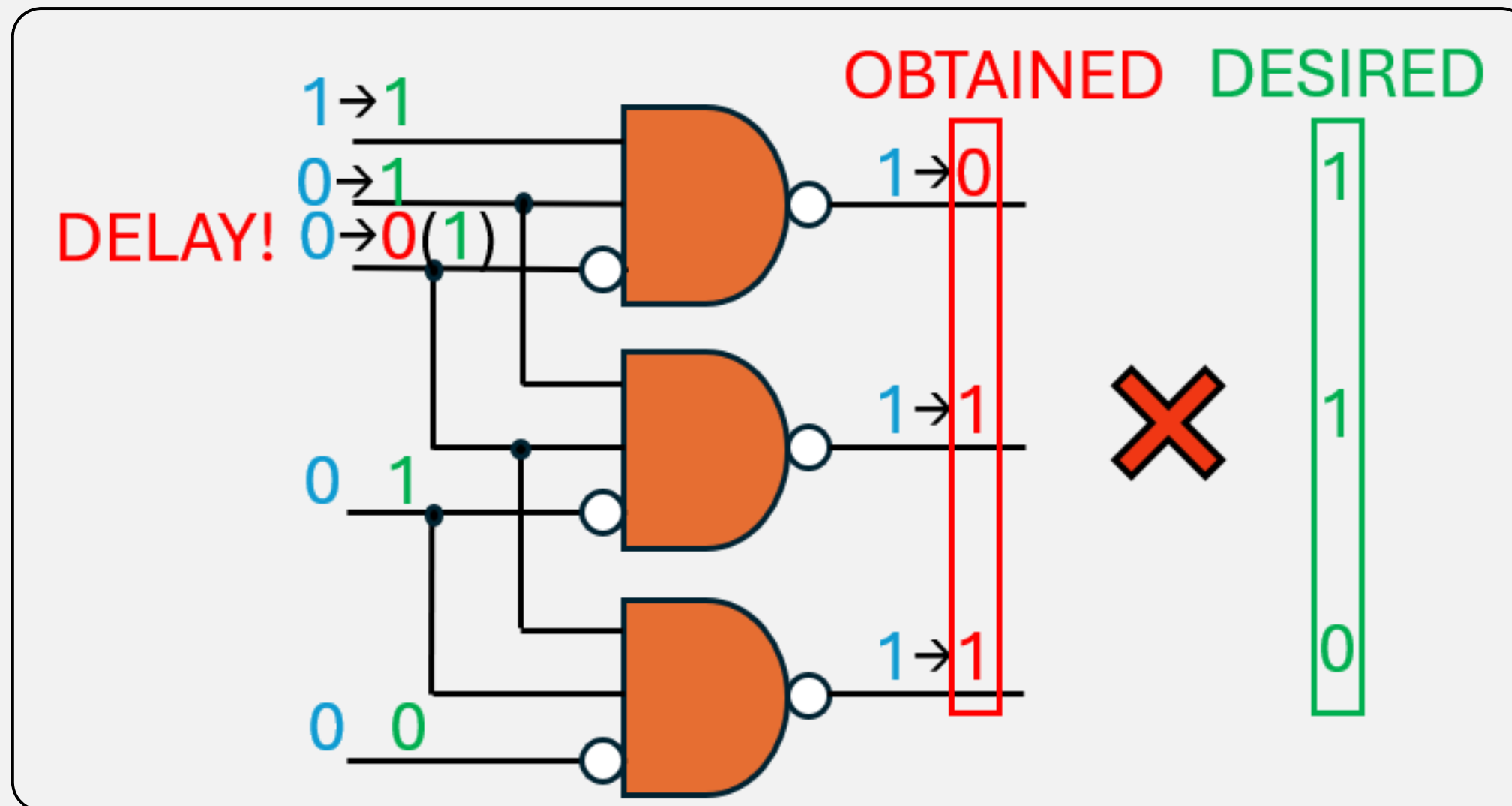
Type “0” bubbles (order >1):

- Solved by the encoder



Encoder's
reading
direction

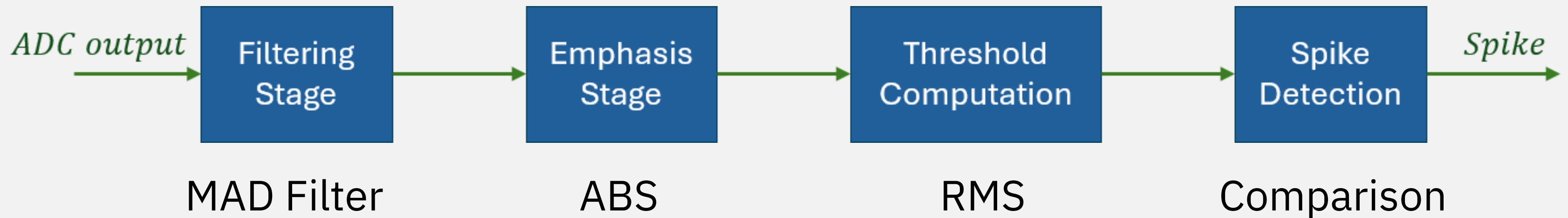
Unsolvable Bubbles



1st order type “0” are the solely bubbles that cannot be solved by this system configuration

- the first NAND3 erroneously decode the input
- the Encoder cannot see any “true” OHE zero

Digital Pre-Processing



Three configurations were considered:

- Base Version (BASE)
- Optimized Version (OPT)
 - Truncated Version (OPT_T2)
- Output Aware Version (EXT)
 - Truncated Version (EXT_T2)

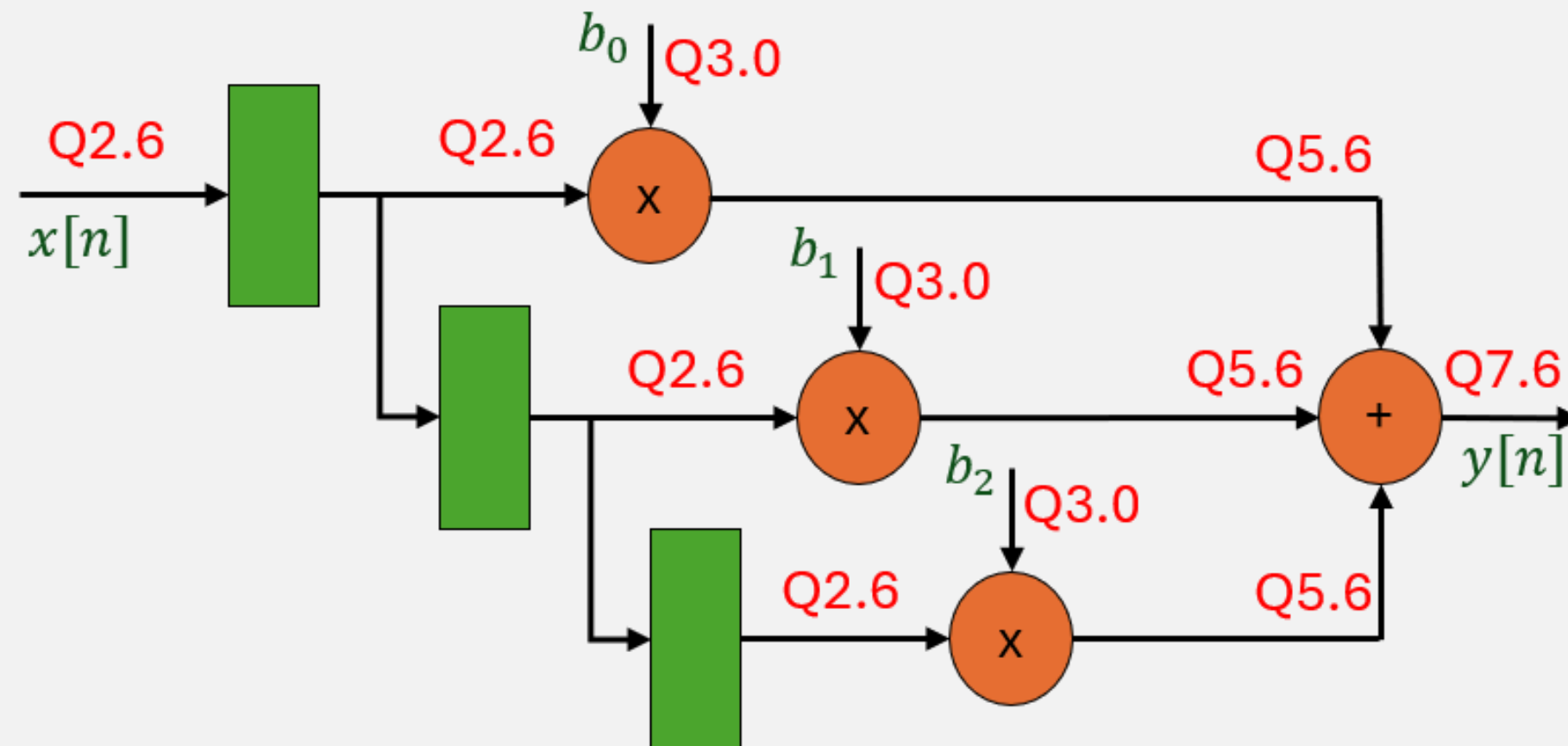
Constraints:

- Throughput of 100 channel x 24 kHz

Filtering Stage: MAD Filter

High-pass filtering:

- Removes 0-400 Hz frequency band



Structure

~~Folded~~

Parallel

~~Parallel L=3~~

Achieved the lowest power consumption

$$y[n] = \sum_{k=0}^2 b_k \times x[n - k]$$

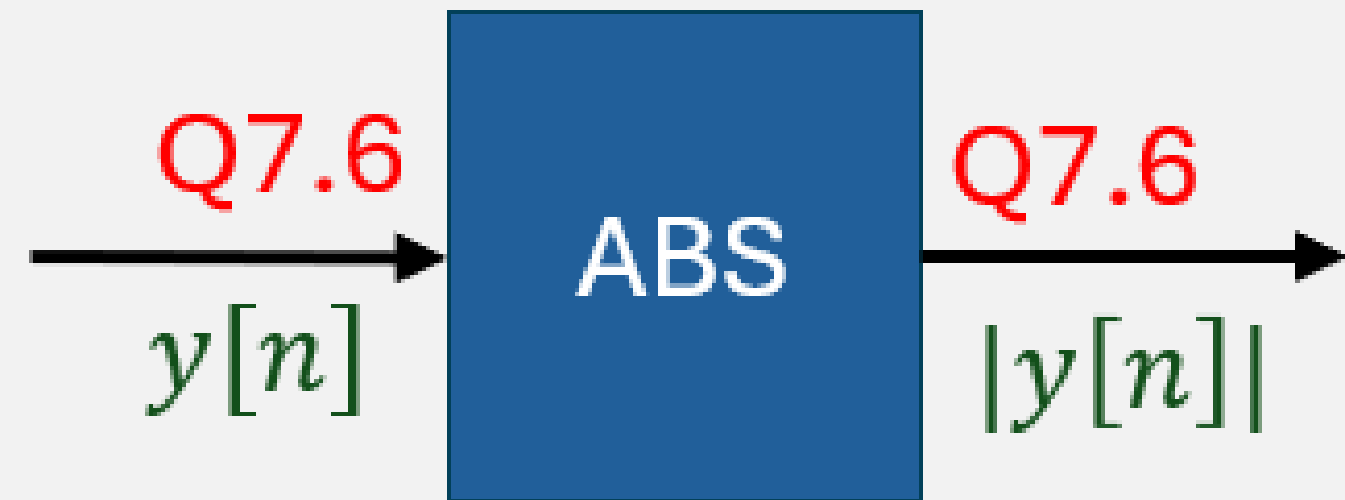
Emphasis Stage: ABS

ABS:

- Absolute value applied to the input

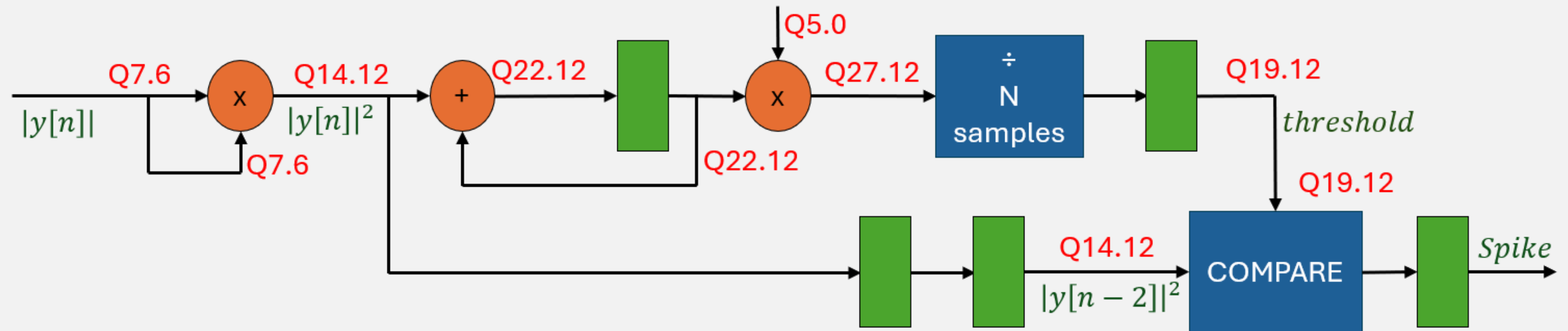
Scope:

- Make the spike waveform more distinguishable from the noise



$$y[n] = |y[n]|$$

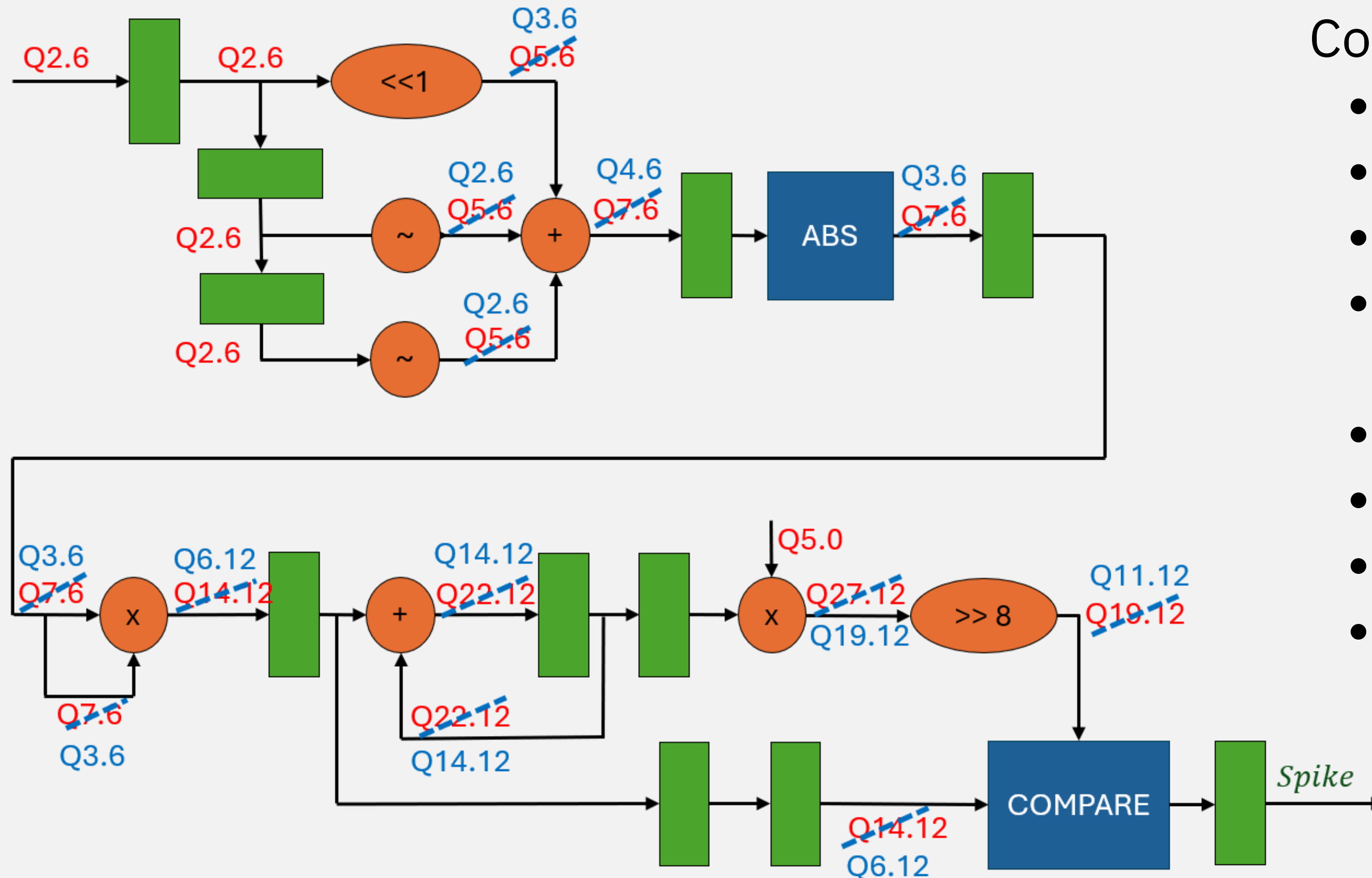
Threshold & Spikes



- Threshold $\rightarrow$ RMS^2 value
- Multiplied by a corrective coefficient
- Computed in a window based manner (N samples)

$$threshold = RMS_{coeff} \times \frac{1}{N} \sum_{i=0}^{N-1} |y_i|^2$$

Optimized Version (OPT)

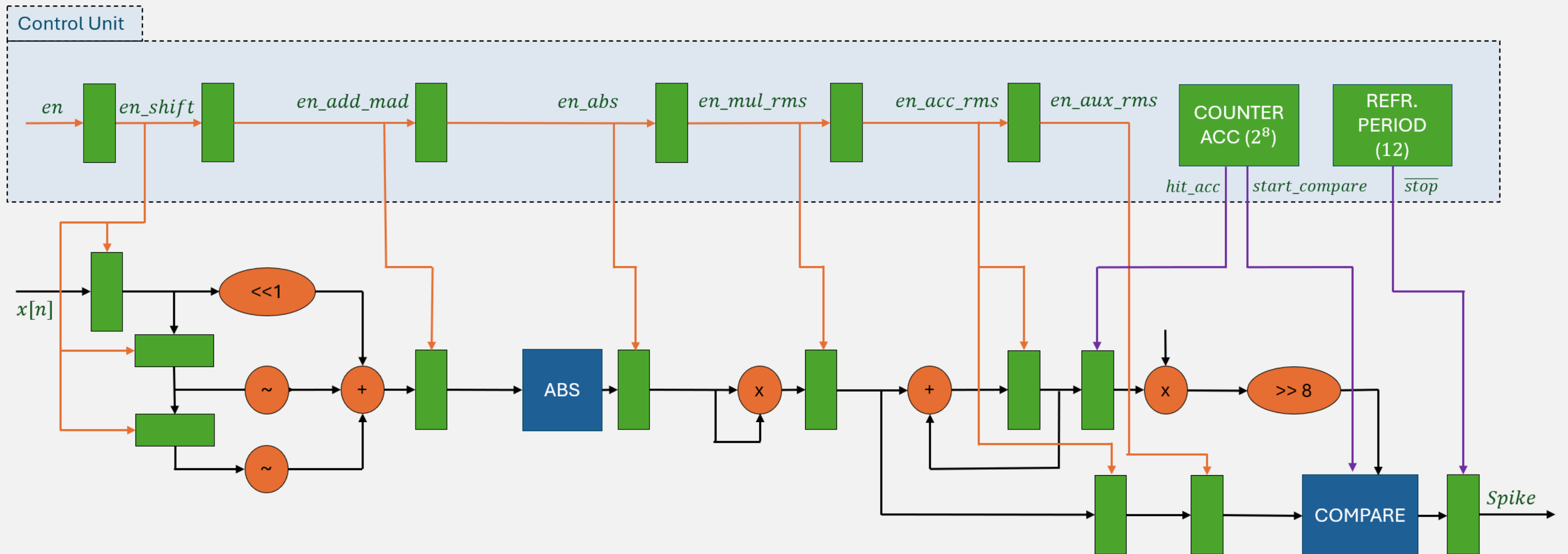


Considering:

- $x[n] \in [-127; 127]$
 - Fixed weights
 - Sign bit removed
 - Shifter implementation
- +
- Clock Gating
 - Pipelining
 - Voltage Scaling
 - Registered Outputs

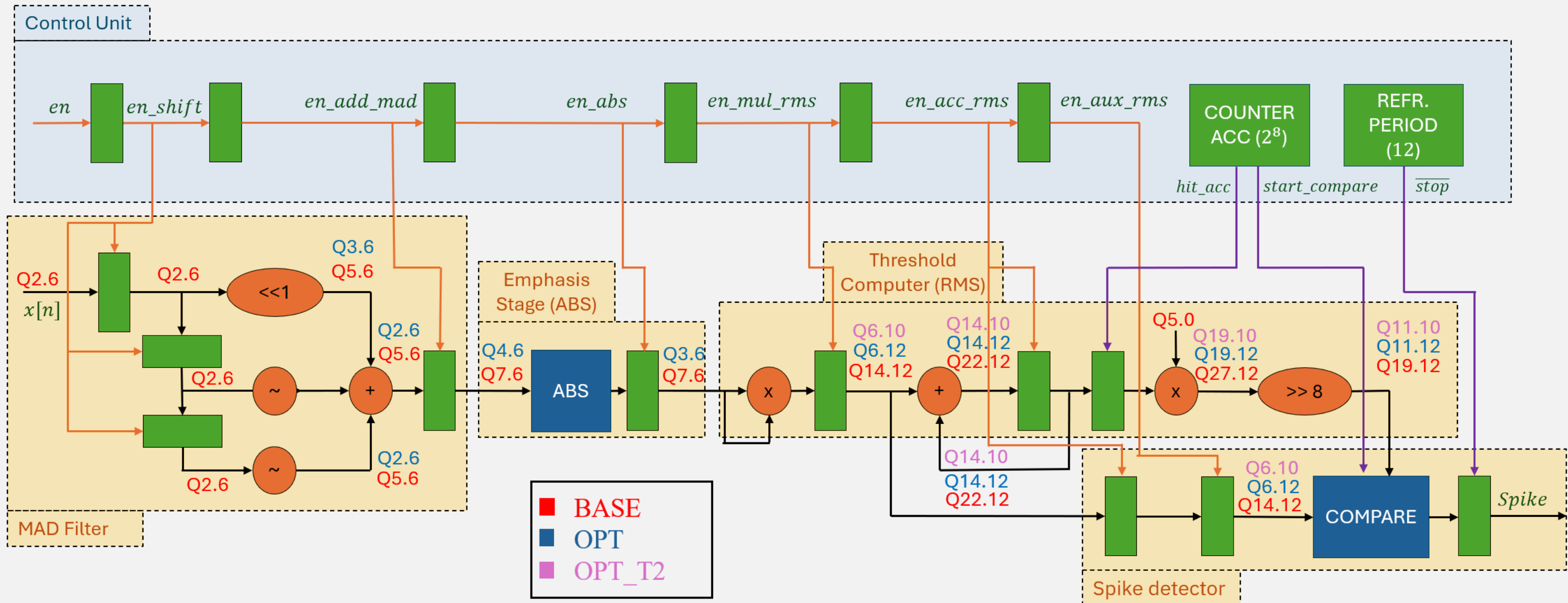
Pipelining - Control Unit

Control Unit: propagates the enable signal through every pipeline stage and generates the control signals

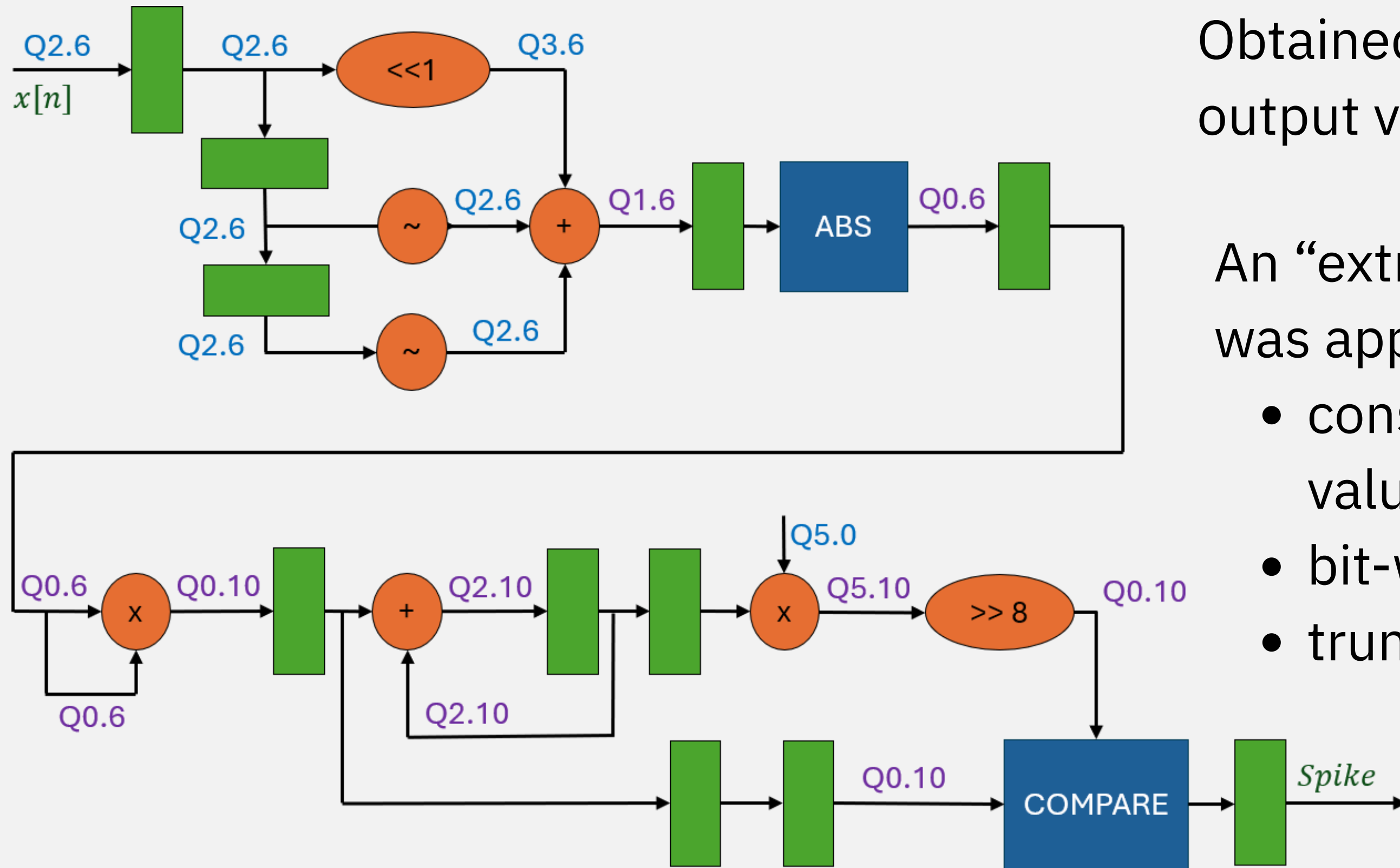


Truncation (OPT_T2)

A truncation of 2 LSB in the “RMS” and “Spike Detector” modules was performed



Output Aware Version (EXT_T2)



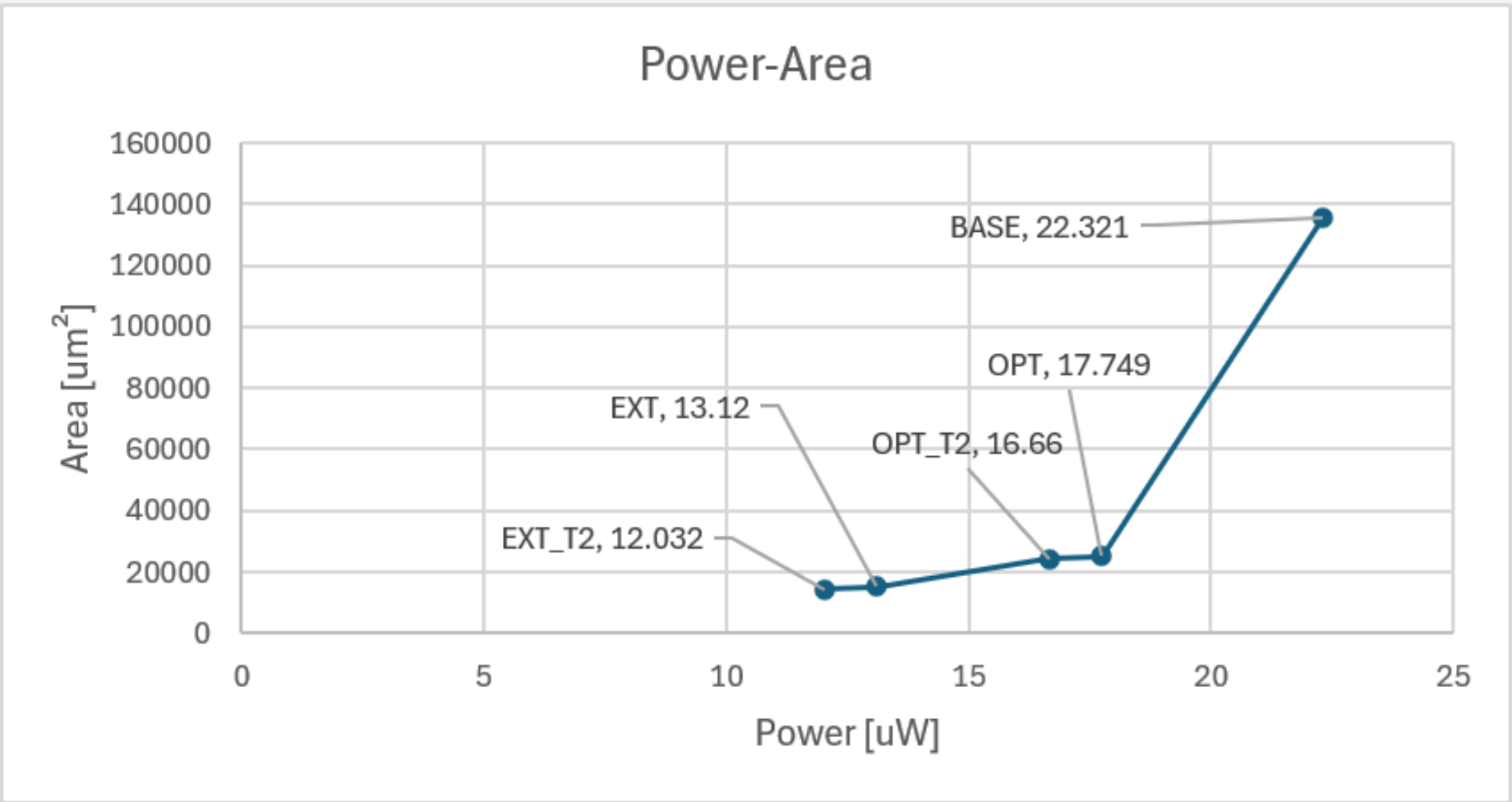
Obtained by looking at the output values at each stage.

An “extreme” optimization was applied:

- considered 10s input values
- bit-width reduction
- truncation performed

Synthesis Comparison

	Power [uW]	Area [um²]	Timing [ps]
BASE	22,321	135705	381060
OPT	17,749	24982	385334
OPT_T2	16,660	24199	388372
EXT	13,120	15292	395724
EXT_T2	12,032	14370	398390



Worst-Case Version BASE

No optimizations
performed

Optimized Version OPT, OPT_T2

Architectural
optimizations and low
power techniques
applied

Output Aware Version EXT - EXT_T2

Optimized version
dimensioned while looking
at the output values

Mixed Signal Simulation and Results

